第 7 次作业解答

L01.08

光行走的时间为

$$t=\int_{\beta_1}^{\beta_2}\frac{n\mathrm{d}s\left(\beta\right)}{c}=\frac{n_0y_0}{c}\int_{\beta_1}^{\beta_2}\frac{\sqrt{\dot{x}^2+\dot{y}^2}}{y}\mathrm{d}\beta.$$

时间取得极值 $\delta t=0$ 的充分必要条件是

$$\frac{\mathrm{d}}{\mathrm{d}\beta}\frac{\partial F}{\partial\dot{x}}-\frac{\partial F}{\partial x}=0,\qquad F=\frac{\sqrt{\dot{x}^2+\dot{y}^2}}{y},$$

关于 y 的方程可以由上式导出, 因为 F 是关于 $\dot{x},\dot{y}$ 的其一次方程. 因为 F 不显含 $\dot{x}$, 所以 x 是可遗坐标, 因此

$$\frac{\partial F}{\partial\dot{x}}=C,\qquad\frac{\dot{x}}{y}\frac{1}{\sqrt{\dot{x}^2+\dot{y}^2}}=C.$$

解得

$$\left(x-C_1\right)^2+y^2=C_2^2,$$

C_1,C_2 是待定常数, 根据光的出射位置和方向决定.

L08.05

曲面面积的表达式是

$$S=\int_{\beta_1}^{\beta_2}2\pi y\mathrm{d}s\left(\beta\right)=2\pi\int_{\beta_1}^{\beta_2}y\sqrt{\dot{x}^2+\dot{y}^2}\mathrm{d}\beta,$$

曲面面积取极值 $\delta S=0$ 的充分必要条件是

$$\frac{\mathrm{d}}{\mathrm{d}\beta}\frac{\partial F}{\partial\dot{x}}-\frac{\partial F}{\partial x}=0,\qquad F=y\sqrt{\dot{x}^2+\dot{y}^2},$$

关于 y 的方程可以由上式导出, 因为 F 是关于 $\dot{x},\dot{y}$ 的其一次方程. 因为 F 不显含 $\dot{x}$, 所以 x 是可

遗坐标, 因此

$$\frac{\partial F}{\partial \dot{x}} = C, \qquad \frac{y\dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = C.$$

解得

$$y = \frac{1}{C_1}\mathrm{sh}\,(C_1 x + C_2),$$

C_1, C_2 是待定常数, 根据点的位置决定.

J05.06

(A)

The arc length derived from every possible curve is expressed as

$$S = \int_{\beta_1}^{\beta_2} \mathrm{d}s\,(\beta) = \int_{\beta_1}^{\beta_2} \sqrt{\dot{r}^2 + r^2\dot{\theta}^2 + \dot{z}^2}\mathrm{d}\beta = \int_{\beta_1}^{\beta_2} \sqrt{a^2\dot{\theta}^2 + \dot{z}^2}\mathrm{d}\beta.$$

The sufficent and neccersary condition of $\delta S = 0$ is

$$\begin{cases} \dfrac{\mathrm{d}}{\mathrm{d}\beta}\dfrac{\partial F}{\partial \dot{\theta}} - \dfrac{\partial F}{\partial \theta} = 0 \\ \dfrac{\mathrm{d}}{\mathrm{d}\beta}\dfrac{\partial F}{\partial \dot{z}} - \dfrac{\partial F}{\partial z} = 0 \end{cases}, \qquad F = \sqrt{a^2\dot{\theta}^2 + \dot{z}^2}.$$

Notice that the θ and z are both ignorable coordinates, so

$$\begin{cases} \dfrac{\partial F}{\partial \dot{\theta}} = C_1 \\ \dfrac{\partial F}{\partial \dot{z}} = C_2 \end{cases}, \qquad \begin{cases} \dfrac{a^2\dot{\theta}}{\sqrt{a^2\dot{\theta}^2 + \dot{z}^2}} = C_1 \\ \dfrac{\dot{z}}{\sqrt{a^2\dot{\theta}^2 + \dot{z}^2}} = C_2 \end{cases}.$$

(B)

If we choose $\beta = s$, we have

$$\begin{cases} a^2\dot{\theta} = C_1 \\ \dot{z} = C_2 \end{cases}.$$

As a result,

$$\begin{cases} \theta = \dfrac{C_1}{a^2} s + \theta_0 \\ z = C_2 s + z_0 \end{cases}.$$

The argument θ_0 and z_0 are determinated by start point's position, the C_1 and C_2 is based on the end point.

(C)

First, let $\theta_0 = 0$ and $z_0 = 0$, and we are going to figure out C_1 and C_2. Let $\theta = 2\pi$ and $z = h$ and $s = \sqrt{(2\pi a)^2 + h^2}$ we have the equation on cylinder surface.

$$\begin{cases} C_1 = \dfrac{2\pi a^2}{\sqrt{(2\pi a)^2 + h^2}} \\ C_2 = \dfrac{h}{\sqrt{(2\pi a)^2 + h^2}} \end{cases}.$$

Then, we consider plattened suiface, let $x = a\theta$ and the equation on the plane is

$$z = \frac{h}{2\pi a} x = \frac{h}{2\pi} \theta,$$

that is definately the same equation in cylinder.

J05.09

(1)

We can easily write down the work of gravity as

$$W_G = \frac{GMm}{2R^3} (R_e^2 - r^2),$$

so we have

$$t = \int_{\beta_1}^{\beta_2} \frac{\mathrm{d}s\,(\beta)}{v} = \int_{\beta_1}^{\beta_2} \frac{\sqrt{\dot{x}^2 + \dot{y}^2}\mathrm{d}\beta}{\sqrt{\dfrac{GM}{R^3}(R_e^2 - r^2)}} = \sqrt{\frac{R^3}{GM}} \int_{\beta_1}^{\beta_2} \sqrt{\frac{\dot{x}^2 + \dot{y}^2}{R_e^2 - r^2}} \mathrm{d}\beta.$$

The Euler-Lagrange equations are

$$\frac{\mathrm{d}}{\mathrm{d}\beta} \frac{\partial F}{\partial \dot{x}} - \frac{\partial F}{\partial x} = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}\beta} \frac{\partial F}{\partial \dot{y}} - \frac{\partial F}{\partial y} = 0, \qquad F = \sqrt{\frac{\dot{x}^2 + \dot{y}^2}{R_e^2 - r^2}}, \qquad r^2 = x^2 + y^2.$$

Expand them,

$$\frac{\mathrm{d}}{\mathrm{d}\beta} \left(\frac{\dot{x}}{\sqrt{(R_e^2 - x^2 - y^2)(\dot{x}^2 + \dot{y}^2)}} \right) - \sqrt{\frac{\dot{x}^2 + \dot{y}^2}{R_e^2 - x^2 - y^2}} \frac{x}{R_e^2 - x^2 - y^2} = 0,$$

$$\frac{\mathrm{d}}{\mathrm{d}\beta}\left(\frac{\dot{y}}{\sqrt{\left(R_e^2-x^2-y^2\right)\left(\dot{x}^2+\dot{y}^2\right)}}\right)-\sqrt{\frac{\dot{x}^2+\dot{y}^2}{R_e^2-x^2-y^2}}\frac{y}{R_e^2-x^2-y^2}=0.$$

(2)

We can verify that

$$\begin{cases}\beta=\eta\\ x=(R-a)\cos\eta+a\cos\left(\dfrac{R-a}{a}\eta\right)\\ y=(R-a)\sin\eta-a\sin\left(\dfrac{R-a}{a}\eta\right)\end{cases}$$

satisfy the Euler-Lagrange equation if $R=R_e$.

(3)

Let

$$x^2+y^2=R_e^2,$$

and $\eta=\frac{2\pi a}{R}$. So $S=2\pi a=D$ and $a=\frac{D}{2\pi}$, what more

$$h_{\max}=R_e-\sqrt{\min\left\{x^2+y^2\right\}}=2a=\frac{D}{\pi}.$$

Based on it

$$t=\sqrt{\frac{{R_e}^3}{GM}}\int_0^{2a\pi/R}\sqrt{\frac{\dot{x}^2+\dot{y}^2}{R_e^2-r^2}}\mathrm{d}\eta=\sqrt{\frac{{R_e}^3}{GM}}\frac{D}{R}\sqrt{\frac{2\pi R}{D}-1}.$$

(4)

$$\frac{d}{ds}\left(\frac{\hat{\boldsymbol{\tau}}}{\sqrt{R^2-r^2}}\right)-\frac{\boldsymbol{r}}{(R_e^2-r^2)^{3/2}}=0.$$

Here we have

$$\sqrt{\dot{x}^2+\dot{y}^2}=1,\qquad \dot{x}\hat{\boldsymbol{x}}+\dot{y}\hat{\boldsymbol{y}}=\hat{\tau}.$$